SharePoint Online Search & Information Architecture

Implementation Solution Guide

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Executive Summary — SharePoint Online Search & Information Architecture

Outcome. Content in SharePoint Online becomes reliably findable: the right document or page surfaces on the first page of results, users can filter by meaningful attributes (department, document type), a dedicated tab exists for high-value categories, and known-item searches jump straight to the authoritative answer. Fewer "I can't find it" tickets, less reliance on emailed copies, and measurably faster time-to-content.

Approach. Findability is fixed in the correct order — **information architecture first, search schema second, search experience third** — and delivered with the house model **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**:

- **Metadata & content types:** a small, governed set of term sets and enterprise content types (`Content Type – Policy Document – Prod`) so content is described consistently tenant-wide.
- **Search schema:** crawled properties mapped to reusable refinable managed properties with friendly aliases (`Department`, `DocType`) and the right searchable/queryable/retrievable/refinable flags.
- **Search experience:** scoped **result sources**, a **vertical** (a Policies tab), and **promoted results / bookmarks** for known-item queries — configured on the Microsoft Search surface where users actually start.
- **Automation:** five PowerShell tools (SharePoint Online Management Shell / PnP / Graph) for assessment, schema mapping, source/vertical/bookmark configuration, findability validation, and ongoing health.

Why it is low-risk. Everything is proven on a named **pilot** scope with explicit pass/fail gates before any tenant-wide change. Every tenant-level schema change is preceded by a captured **rollback baseline** (a full search-config export) and made in a change window. The scripts are **safe-by-default** — read-only or `-WhatIf` unless an explicit commit switch is passed, modern-auth only, no stored credentials. Search reflects existing permissions (it never grants access), and all compliance/records/legal determinations remain with those functions. Plan- and cloud-dependent behavior (content-type gallery vs hub, Microsoft Search availability, Government-cloud URLs) is flagged to verify in-tenant rather than assumed.

What you receive. A solution overview, a requirements guide, an implementation runbook with named artifacts and gates, testing and troubleshooting guides, best practices, and the five PowerShell tools — enough for a competent M365 administrator to run the engagement end to end for a fraction of the cost of bespoke consulting.

1. Solution Overview — SharePoint Online Search & Information Architecture

What this solution is

When users say "I can't find anything in SharePoint," the fix is almost never "buy more SharePoint" — it is *search tuning on a sound information architecture (IA)*. Microsoft Search and classic SharePoint search

only surface what the content is *described* as: the metadata on it, the content types that carry that metadata, and the **search schema** that decides which of that metadata is searchable, queryable, retrievable, and refinable. This solution is a repeatable method to establish the metadata and content-type layer, tune the search schema and managed properties, stand up result sources, verticals, and promoted results, and validate that the *right* content surfaces fast — using the SharePoint admin center, the Search admin center, managed metadata, and PowerShell (SharePoint Online Management Shell / PnP PowerShell / Microsoft Graph). No third-party search product is required.

Why this matters

• Search is the single most-used feature in SharePoint, and the first thing users blame. Poor

findability quietly pushes people back to email attachments and local copies — the failure mode the intranet was meant to end.

• Search quality is *downstream of IA*. If content has no consistent metadata (department, document

type, owner, status) there is nothing for the schema to refine or filter on, so results stay a flat, relevance-only list. Metadata first, schema second, experience third.

• The knobs that actually move findability — **managed property mappings**, **searchable / queryable /

retrievable / refinable **flags,** result sources**,** verticals**, and** promoted results / bookmarks** — are tenant- and site-collection-level settings that are easy to get wrong and, once wrong, silently degrade every query. This solution makes those changes deliberate, named, piloted, and reversible.

• A one-time "fix search" project regresses as new sites, libraries, and content types appear without

metadata discipline. The durable value is the schema + governance pattern, not a single sweep.

How it works

1. **Crawling** — SharePoint continuously crawls tenant content. Each column/field on an item becomes a

crawled property (e.g. a `Department` site column surfaces as a crawled property like `ows_Department`). Crawled properties are raw and, by default, not individually queryable or refinable.

1. **The search schema** maps **crawled properties** → **managed properties**. A managed property is the

named, typed handle search actually uses. Each managed property has attributes:

• **Searchable** — its text is included in the full-text index (matches a free-text query).

• **Queryable** — it can be targeted by name in a query (`Department:Finance`).

• **Retrievable** — its value can be returned in results (shown in a card / custom result).

• **Refinable** — it can drive a refiner/filter on a results page (sort/filter by it).

• **Sortable** — results can be ordered by it.

Out of the box, tenant-level managed properties are read-only; you map metadata onto the built-in, reusable `RefinableString00…`**,** `RefinableDate00…`**,** `RefinableInt00…` aliases (and set an **alias** so queries use a friendly name).

1. **Managed metadata & content types** feed the schema. A **term set** (e.g. Department, Document Type)

defines the controlled vocabulary; **site columns** bound to those term sets carry it; **content types** bundle those columns so every library applies them consistently. Published from a central content-type gallery / hub, they give the whole tenant the same crawled properties to map.

1. **Result sources** scope *where* results come from (e.g. only a Policies hub, only documents of a

given content type) using a query transform. **Verticals** are the tabs on the Microsoft Search / results page (All, Files, Sites, plus custom ones like "Policies") — each backed by a result source.

1. **Promoted results / bookmarks** pin an authoritative answer to the top for known queries (e.g.

searching "VPN" always surfaces the IT VPN page). In Microsoft Search these are **Bookmarks**; classic query rules can also promote results within a site.

1. **The Microsoft Search experience** (the search box in the M365 app bar, Office.com, and SharePoint

home) presents verticals, bookmarks, acronyms, and Q&A tenant-wide. This solution tunes that surface so answers appear where users start.

1. **Validation** runs representative **test queries** and confirms the target content is found, the

managed property is returned/refinable, and the vertical and bookmark behave — before expanding tenant-wide.

The reference design

- A central **content-type gallery** publishes a small, governed set of enterprise content types

(`Content Type – Policy Document – Prod`, `Content Type – Contract – Prod`) each carrying term-set-bound site columns (`Site Column – Department`, `Site Column – DocType`, `Site Column – Owner`).

- Those site columns surface as crawled properties, mapped to refinable managed properties with friendly

aliases (`RefinableString00` → alias `Department`, `RefinableString01` → alias `DocType`).

- A **Policies result source** and a **Policies vertical** give users a dedicated, scoped tab; **bookmarks**

pin authoritative answers for high-volume known-item queries.

- Everything is proven on a **pilot hub / site set** with pass/fail gates and a documented rollback

(schema export, source/vertical/bookmark removal) before tenant-wide expansion.

What you can control

- Which term sets and content types are in scope (start with the highest-value: policies, contracts, HR).
- Which crawled properties map to which refinable managed properties, and their alias names.
- Which result sources and verticals exist, and the query transform that scopes each.
- Which known-item queries get promoted results / bookmarks.

• The pilot scope validated first, before any tenant-wide schema or Microsoft Search change.

Limitations (be honest with buyers)

• **Reindex latency.** Changing a managed property mapping does not retro-apply until the affected content

is **re-crawled/re-indexed**. You can force a reindex per library/site (Advanced settings → Reindex), but tenant-scale reindex takes time — plan for hours-to-days, not minutes, before refiners populate.

• **Refinable managed properties are a finite, typed pool** (`RefinableString00–199`, `RefinableDate`,

`RefinableInt`, etc.). Track allocations so two teams don't map different metadata to the same alias.

• **Tenant-level schema is global.** A mapping made at the tenant Search admin center affects every site;

site-collection-level schema (`/_layouts/15/listmanagedproperties.aspx`) is narrower but only overrides where allowed. Change tenant-level schema under change control.

• **Managed metadata / content-type publishing** (content-type gallery vs the legacy Content Type Hub) is

plan- and rollout-dependent — confirm which your tenant exposes; the steps flag "verify in your tenant" where Microsoft is mid-transition.

• **Government clouds (GCC High / DoD)** use different portal URLs (`*.sharepoint.us` /

`*-admin.sharepoint.us`, `*.office365.us`) and some Microsoft Search features roll out later — verify availability in your cloud.

• **Compliance/records determinations** (which content is a record, retention labels, information barriers)

are **deferred** to your compliance/records/legal functions — this solution makes content *findable*, it does not decide what content must be *retained or restricted*.

Naming convention (used in every doc and script)

Every artifact this solution creates is named `<Purpose> – <Scope> – <Stage>` and reused verbatim across the docs, scripts, admin centers, and search schema.

Artifact	Name
Department term set	`Term Set – Department – Prod`
Document-type term set	`Term Set – DocType – Prod`
Department site column	`Site Column – Department`
Document-type site column	`Site Column – DocType`
Enterprise policy content type	`Content Type – Policy Document – Prod`
Enterprise contract content type	`Content Type – Contract – Prod`
Managed property alias (Department)	`Department` (on `RefinableString00`)

Artifact	Name
Managed property alias (DocType)	`DocType` (on `RefinableString01`)
Policies result source	`Result Source – Policies – Prod`
Policies search vertical	`Search Vertical – Policies – Prod`
Known-item bookmark	`Bookmark – VPN Access – Prod`
Pilot site set	`Pilot Site – Findability – Pilot`
Findability assessment export	`Findability Assessment – Tenant – Discovery`
Findability validation report	`Findability Validation – Pilot – Validation`

The standard deployment model

> *Assess current metadata/schema/findability → design the IA + schema map → configure metadata, content types, managed properties, sources/verticals/bookmarks → prove it on a **pilot** hub/site → validate with test queries against pass/fail gates → expand tenant-wide. Pattern: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, pilot-first, with a documented rollback for every tenant-level change.*

2. Requirements — SharePoint Online Search & Information Architecture

These are the **solution-specific** prerequisites for tuning SharePoint Online search and information architecture. They build **on top of** the shared SharePoint-category prerequisites in `global-requirements-sharepoint.md` (M365 tenant + SharePoint/OneDrive licensing; least-privilege SharePoint admin access; SPO Management Shell / PnP / Graph toolchain; an IA & identity source of truth; a pilot scope + change control; and security/compliance/records stakeholders). Verify those six first, then confirm the search-and-IA–specific items below. Each: **Description / Why / Verify / Implement if missing.**

1. Search Administrator access (Search admin center)

• **Description:** An account with **Search Administrator** (or SharePoint Administrator / Global Administrator, which include it) able to open the **Microsoft Search & Intelligence admin center** (`admin.microsoft.com` → Settings → Search & intelligence) and the SharePoint **Search admin center** (`https://<tenant>-admin.sharepoint.com` → More features → Search).

• **Why required:** Tenant-level managed properties, result sources, verticals, and Microsoft Search bookmarks/verticals are configured only with the search-admin surfaces.

• **How to verify:** Sign in and confirm the "Search & intelligence" node and the SPO admin "Search"

page load without an access error. In Government clouds the URLs are `*-admin.sharepoint.us` / `admin.microsoft.us` — **verify in your tenant**.

- **How to implement if missing:** Have a Global Admin assign **Search Administrator** (Entra role) under

change control; use the least-privilege role that covers the scope you are changing.

2. Managed metadata service & term-store contributor rights

- **Description:** The **Managed Metadata Service** application (term store) is provisioned, and you have

Term Store Administrator (or Group Manager/Contributor on the relevant term group) to create the Department and Document-Type term sets.

- **Why required:** The controlled vocabularies (`Term Set – Department – Prod`, `Term Set – DocType – Prod`)

that drive consistent metadata — and therefore refinable managed properties — live in the term store.

- **How to verify:** Open the SharePoint admin center → **Content services** → **Term store**; confirm the

term store loads and your account appears as an admin/contributor.

- **How to implement if missing:** In Term store, add your account as a Term Store Administrator (or the

taxonomy owner grants Group Manager on the working term group).

3. Content-type publishing surface (gallery / hub) confirmed

- **Description:** The mechanism your tenant uses to publish enterprise content types across sites —

the modern **content type gallery** (SharePoint admin center → Content services → Content type gallery) and/or the legacy **Content Type Hub** (`/sites/contentTypeHub`).

- **Why required:** Enterprise content types (`Content Type – Policy Document – Prod`,

`Content Type – Contract – Prod`) must publish tenant-wide so every library produces the same crawled properties to map.

- **How to verify:** In the SharePoint admin center confirm **Content type gallery** is present; check

whether a legacy Content Type Hub site also exists. Microsoft is mid-transition between the two — **verify which your tenant exposes** before authoring content types.

- **How to implement if missing:** Use the content type gallery if present; otherwise enable/locate the

Content Type Hub. Decide one publishing path and record it before creating content types.

4. A defined findability intent (the IA + query inventory)

- **Description:** A short, agreed statement of **what users need to find** and **how they search for it**:

the top content domains (policies, contracts, HR, projects), the metadata that describes them, and a list of representative **known-item and exploratory test queries**.

- **Why required:** Managed-property mappings, result sources, verticals, and bookmarks are only "right"

relative to real queries; the test-query inventory is also the validation instrument (§5 Testing).

- **How to verify:** Confirm a documented list of priority content domains, the metadata each carries, and 10–20 representative test queries with the expected top result for each.
- **How to implement if missing:** Interview content owners and the service desk (top search terms, top "I can't find X" tickets); capture the query inventory before configuring. This is a design input, not a compliance determination.

5. Refinable managed-property allocation register

- **Description:** A simple register of which reusable **refinable** managed properties (`RefinableString00–199`, `RefinableDate00`…, `RefinableInt00`…, `RefinableDecimal00`…) are already in use and which are free.
- **Why required:** The refinable pool is finite and typed; mapping two different site columns to the same alias corrupts refiners tenant-wide. A register prevents collisions.
- **How to verify:** Export the current search schema (`Get-PnPSearchConfiguration -Scope Subscription`, or the Search admin center "Managed Properties" list) and record which refinables already carry a mapping/alias.
- **How to implement if missing:** Run the discovery script (`Get-SearchIAAssessment.ps1`) to enumerate current managed properties and their mappings; seed the register from its output.

6. A pilot hub/site set with representative content

- **Description:** A **named pilot scope** — a hub or a small set of site collections (`Pilot Site – Findability – Pilot`) that contains real, representative content and can absorb a forced reindex without disrupting the tenant.
- **Why required:** Schema, source/vertical, and bookmark changes must be proven on real content and re-indexed there first; a pilot lets you validate refiners populate and queries return correctly before tenant-wide schema changes (which reindex everything).
- **How to verify:** Confirm the pilot sites exist, carry the target content types/metadata, and that a reindex there is acceptable in the change window.
- **How to implement if missing:** Designate the pilot sites, apply the content types/metadata to their key libraries, and agree the change window before configuring the schema.

7. Reindex / change window and rollback baseline

- **Description:** An agreed **change window** for tenant-level schema changes (which trigger a full

reindex over time) and a captured **rollback baseline** — an export of the current search configuration before any change.

- **Why required:** Tenant-level managed-property and result-source changes are high-impact and take time to

propagate; you must be able to revert to the pre-change schema.

- **How to verify:** Confirm `Get-PnPSearchConfiguration -Scope Subscription -Path .\SearchConfig-Baseline.xml`

produces a baseline file and it is stored with the change record.

- **How to implement if missing:** Capture the baseline export (the discovery/health scripts do this) and

schedule the change window before applying tenant-level schema changes.

Global Requirements — SharePoint Online & OneDrive Solutions

These requirements apply to **every** solution in the **SharePoint** category (SharePoint Online architecture & provisioning, external sharing, OneDrive for Business, migration to SPO, permissions, search, and records/retention). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every SharePoint solution).*** *The **SharePoint admin center**, the **Microsoft 365 admin center**, the **SharePoint Online Management Shell** and **PnP PowerShell**, and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Where a capability, setting, or portal path is **plan- or version-dependent** (storage limits, sharing controls, Syntex/records features, Commercial vs Government cloud), it is **cited or flagged "verify in your tenant"** — never invented. **Third-party migration and governance tools** (e.g. ShareGate, Migration Manager, AvePoint) are integrated at the pattern level; their consoles are deferred to the vendor's docs. Automation is **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with SharePoint Online / OneDrive and licensing

- **Description:** A Microsoft 365 tenant with **SharePoint Online** and **OneDrive for Business** and

the licensing the capability requires (e.g. an E-plan for advanced records/retention, Syntex where used).

- **Why required:** All site, sharing, OneDrive, and records configuration depends on the right plan.
- **How to verify:** In the M365 admin center confirm the SharePoint/OneDrive subscription and add-ons.
- **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Administrative access (least-privilege SharePoint roles)

- **Description:** Access with the least-privilege role the solution names — typically **SharePoint

Administrator** (and Global Administrator only where genuinely required, e.g. tenant-wide sharing).

- **Why required:** Tenant/site settings, sharing, provisioning, and records changes require the role.

- **How to verify:** Confirm the role in the M365 admin center / Entra roles.
- **How to implement if missing:** Have a Global Admin assign the least-privilege role under change control.

3. SharePoint Online Management Shell / PnP + Graph toolchain

- **Description:** A workstation with the **SharePoint Online Management Shell** and/or **PnP PowerShell** (and Graph PowerShell where used), modern-auth / MFA-capable admin sign-in (PnP app registration where required).
- **Why required:** Configuration, automation, and the provided scripts run through SPO/PnP/Graph.
- **How to verify:** `Connect-SPOService` / `Connect-PnPOnline` succeeds; modules installed.
- **How to implement if missing:** Install the modules; register the PnP app where needed; confirm sign-in.

4. Information architecture & identity/group source of truth

- **Description:** A defined **information-architecture** intent (hub/site structure, taxonomy) and the identity/group source (Entra ID groups, M365 Groups backing team sites) that drives access.
- **Why required:** Provisioning, permissions, search, and governance key off the IA and groups.
- **How to verify:** Confirm the intended hub/site structure and which groups model is authoritative.
- **How to implement if missing:** Define the IA and group model (often the Architecture solution) first.

5. A pilot scope and change control

- **Description:** A **pilot site collection / hub / user set** to validate on before tenant-wide rollout, and a **change-control** process for production changes (sharing, permissions, retention, migration cutover are high-impact).
- **Why required:** Every solution follows an **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** model; tenant-wide sharing/permission/retention changes are high-impact.
- **How to verify:** Confirm a named pilot scope and an approval/change window are available.
- **How to implement if missing:** Create a pilot scope and agree change control before production changes.

6. Security, compliance & records stakeholders

- **Description:** Engagement with the **security, compliance/records, and data-governance** stakeholders whose policies the configuration must satisfy (external sharing, permissions, retention/records, sensitivity).
- **Why required:** Sharing, permission, and records settings must align to enterprise policy.

• **How to verify:** Confirm the applicable policies and named approvers.

• **How to implement if missing:** Identify the owners and capture requirements before configuring governance.

*The deployment model across every SharePoint solution is the same: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes. Automation is delivered as **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets in code.*

3. Implementation — SharePoint Online Search & Information Architecture

Follow the house deployment model: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, pilot-first, with explicit pass/fail gates and a documented rollback. Every artifact is named `<Purpose> – <Scope> – <Stage>` (see Overview) and reused verbatim in later steps, testing, troubleshooting, and the scripts. Navigation is given as both admin-center paths **and** PowerShell.

> ***Connections used throughout.*** *- SharePoint Online Management Shell:* `Connect-SPOService -Url https://<tenant>-admin.sharepoint.com` *- PnP PowerShell (schema, metadata, content types, queries):* `Connect-PnPOnline -Url https://<tenant>-admin.sharepoint.com -Interactive` *(or* `-Url https://<tenant>.sharepoint.com/sites/<site>` *for site-scoped work). - Modern auth / MFA only;* ***no stored credentials****. Government cloud: substitute* `*.sharepoint.us` */* `*-admin.sharepoint.us` *and* `-AzureEnvironment USGovernment` *where applicable —* ***verify in your tenant****.*

Stage 0 — Assess (read-only)

Goal: capture the current metadata, content-type, and search-schema state, and a rollback baseline.

1. **Run the assessment.** From the toolchain workstation:

```
.\Get-SearchIAAssessment.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -OutputPath .\Findability-Assessment-Tenant-Discovery.json
```

This enumerates term groups/term sets, published content types, site columns, and the current managed-property → crawled-property mappings (with searchable/queryable/retrievable/refinable flags). Save the export as `Findability Assessment – Tenant – Discovery`.

1. **Capture the rollback baseline** of the search schema:

```
Get-PnPSearchConfiguration -Scope Subscription -Path .\SearchConfig-Baseline.xml
```

Store `SearchConfig-Baseline.xml` with the change record — this is the tenant-level rollback artifact.

1. **Seed the refinable register** (Requirements §5) from the assessment output: list which

`RefinableString/Date/Int` slots already carry a mapping/alias, and which are free.

Gate 0 (pass/fail): you have (a) a saved assessment, (b) a tenant schema baseline export, and (c) a refinable-allocation register. If any is missing, do not proceed.

Stage 1 — Design

Goal: decide the metadata, the schema map, and the search experience before touching production.

1. **Choose the priority content domains** from the findability intent (Requirements §4) — e.g. Policies, Contracts.
1. **Define the metadata:** two term sets to start — `Term Set – Department – Prod`, `Term Set – DocType – Prod` — plus the site columns `Site Column – Department`, `Site Column – DocType`, `Site Column – Owner`.
1. **Define the content types:** `Content Type – Policy Document – Prod`, `Content Type – Contract – Prod`, each including the three site columns above.
1. **Plan the schema map** using free slots from the register:

Site column (crawled property)	Managed property	Alias	Flags to set
`Site Column – Department` (`ows_Department`)	`RefinableString00`	`Department`	Queryable, Retrievable, Refinable
`Site Column – DocType` (`ows_DocType`)	`RefinableString01`	`DocType`	Queryable, Retrievable, Refinable
`Site Column – Owner` (`ows_Owner`)	`RefinableString02`	`DocOwner`	Queryable, Retrievable

> *Exact crawled-property names appear only **after** the content is first crawled with the column populated (Stage 3.2). Confirm the real `ows_*` name in the assessment before mapping.*

1. **Plan the experience:** `Result Source – Policies – Prod` (scoped to the Policies content), a `Search Vertical – Policies – Prod` backed by it, and the first bookmarks (e.g. `Bookmark – VPN Access – Prod`).
1. **Record the plan** and get change-control approval for the tenant-level steps.

Gate 1: the metadata list, content-type list, schema map (with confirmed free refinable slots), and experience plan are documented and approved.

Stage 2 — Configure metadata & content types

Goal: create the vocabulary and content types so libraries produce consistent crawled properties.

1. **Create the term sets.** SharePoint admin center → **Content services** → **Term store** → your term group → **New term set** → name `Term Set – Department – Prod`; add terms (e.g. Finance, HR, IT, Legal). Repeat for `Term Set – DocType – Prod` (Policy, Contract, Procedure, Report). PowerShell (PnP):

```
Connect-PnPOnline -Url https://<tenant>.sharepoint.com/sites/<taxonomy-site> -Interactive
New-PnPTermGroup -Name "Enterprise IA" -ErrorAction SilentlyContinue
New-PnPTermSet -TermGroup "Enterprise IA" -Name "Term Set – Department – Prod"
Add-PnPTerm -TermGroup "Enterprise IA" -TermSet "Term Set – Department – Prod" -Name "Finance"
```

1. **Create the site columns** as **Managed Metadata** columns bound to the term sets. In the content type

gallery / hub site: **Site settings** → **Site columns** → **Create** → type *Managed Metadata* → `Site Column – Department` → bind to `Term Set – Department – Prod`. Repeat for `Site Column – DocType`. Add `Site Column – Owner` (Person or single line of text). PnP: `Add-PnPField` / `Add-PnPTaxonomyField`.

1. **Create the content types** in the **content type gallery** (or Content Type Hub — **verify which your

tenant uses**):** New content type** → `Content Type – Policy Document – Prod` (parent: Document) → add `Site Column – Department`, `Site Column – DocType`, `Site Column – Owner`. Repeat for `Content Type – Contract – Prod`. PnP: `Add-PnPContentType`, then `Add-PnPFieldToContentType`.

1. **Publish** the content types (content type gallery: **Publish**; Content Type Hub: content type →

Manage publishing → **Publish**) so they propagate to subscribing sites.

Gate 2: term sets exist and are populated; site columns are bound to them; both content types include the three columns and are **published**. Verify with the assessment script or `Get-PnPContentType`.

Stage 3 — Configure the search schema (pilot scope first)

Goal: get real crawled properties and map them to refinable managed properties — proven on the pilot.

1. **Apply content types + populate metadata** on the pilot libraries in `Pilot Site – Findability – Pilot`:

add `Content Type – Policy Document – Prod` to the policies library and tag a representative set of documents with Department/DocType/Owner values.

1. **Trigger a reindex of the pilot library** so the populated columns become crawled properties: library →

Settings → **Library settings** → **Advanced settings** → **Reindex document library**. (Site-scope: Site settings → Search and offline availability → **Reindex site**.) Allow the crawl to complete (minutes to hours).

1. **Confirm the crawled properties exist.** Search admin center → **Manage Search Schema → Crawled

Properties** → search for `Department` / `DocType` and note the exact `ows_*` names. Or re-run `Get-SearchIAAssessment.ps1` and read the crawled-property list.

1. **Map crawled** → **managed properties** with the config helper (dry-run first, then commit within the change

window):

```
# Dry-run (default is -WhatIf; shows the intended mappings/aliases, changes nothing)
.\Set-ManagedPropertyMapping.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -MappingCsv .\schema-map.csv -WhatIf
```

```
# Apply (explicit)
.\Set-ManagedPropertyMapping.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -MappingCsv .\schema-map.csv -Commit
```

`schema-map.csv` lists each crawled property, the target refinable managed property, the alias, and the flags from the Stage-1 map (e.g. `ows_Department,RefinableString00,Department,Queryable;Retrievable;Refinable`). Equivalent manual path: Search admin center → **Manage Search Schema** → **Managed Properties** → open `RefinableString00` → **Add a mapping** to the crawled property → set **Alias** `Department` and check *Queryable / Retrievable / Refinable*.

> *Set the mapping at **site-collection scope** for the pilot first (`/_layouts/15/listmanagedproperties.aspx`) where possible; promote to **tenant scope** only in Stage 6.*

1. **Reindex again** so the new mappings take effect on the pilot content.

Gate 3: for the pilot content, the assessment/validation shows `Department` and `DocType` mapped with Queryable + Retrievable + Refinable set, and their crawled properties identified. If the crawled property never appears, the column was not populated or the reindex has not completed — do not map blindly.

Stage 4 — Configure result sources, verticals & promoted results (pilot)

Goal: give users a scoped Policies tab and pin authoritative answers.

1. **Create the result source** `Result Source – Policies – Prod`. Search admin center →

Manage Result Sources → **New Result Source** → Protocol: *Local SharePoint*, Type: *SharePoint Search Results*, **Query Transform**: `{searchTerms} ContentTypeId:0x0101...* Path:https://<tenant>.sharepoint.com/sites/policies*` (scope by the Policies content type ID and/or the Policies hub path). Or use the config helper:

```
.\Set-SearchSourceVertical.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -ConfigJson .\search-experience.json -WhatIf      # dry-run
.\Set-SearchSourceVertical.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -ConfigJson .\search-experience.json -Commit       # apply
```

1. **Create the vertical** `Search Vertical – Policies – Prod`. **Microsoft Search & Intelligence admin

center → Customizations → Verticals → Add → **name it, set the** content source = SharePoint**, and set a** query **that scopes to policies (e.g.** `ContentType:"Policy Document"` **using the published content type), then** enable** it. This adds a *Policies* tab on the Microsoft Search results page.

> *Classic **site-level** search verticals/result-source scoping differ from **Microsoft Search** verticals; this solution configures the Microsoft Search vertical for the tenant-wide experience and, optionally, a site-scoped result source for a hub's own search box — **confirm which surface your users start from.***

1. **Create promoted results / bookmarks.** Microsoft Search & Intelligence admin center → **Answers →**

Bookmarks → Add → `Bookmark – VPN Access – Prod` → **title, URL (the authoritative IT VPN page), and** keywords **(**`vpn`**,** `remote access`**,** `work from home`**). Set state to** Published **(or Test in the pilot). For a single site's classic search, an equivalent is a** Query Rule → Promoted Result** on that

site.

Gate 4: `Result Source – Policies – Prod` returns only policy content for a test query; the `Search Vertical – Policies – Prod` tab appears and lists policies; `Bookmark – VPN Access – Prod` appears above results for its keywords (in Test/pilot state).

Stage 5 — Validate (pilot)

Goal: prove findability against the test-query inventory before tenant-wide changes.

1. Run the validation script against the pilot:

```
.\Test-Findability.ps1 -SiteUrl https://<tenant>.sharepoint.com/sites/policies `
    -QueryCsv .\test-queries.csv -OutputPath .\Findability-Validation-Pilot-Validation.json
```

`test-queries.csv` holds each query, the expected top result URL, and the managed property to check is returned/refinable. The report is `Findability Validation – Pilot – Validation`.

1. Work the pass/fail table in **§5 Testing**. Every priority known-item query must return its expected

result on the first page; refiners for `Department`/`DocType` must populate; the Policies vertical and the bookmark must behave.

Gate 5 (expansion gate): all priority test queries pass, refiners populate, vertical + bookmark work, and no regression in general search on the pilot. Only then expand. If a gate fails, **roll back** (Stage 7 rollback) or fix and re-validate — do not expand.

Stage 6 — Expand (tenant-wide)

Goal: roll the proven configuration to the tenant under change control.

1. **Promote the schema mappings to tenant scope.** In the change window, apply the same mappings at

tenant scope (Search admin center → Manage Search Schema → Managed Properties, or `Set-ManagedPropertyMapping.ps1 -Scope Subscription -Commit`). This triggers a tenant-wide reindex over time — communicate the propagation window.

1. **Broaden content-type adoption.** Apply the published content types to the remaining priority libraries;

populate metadata (or run a metadata-tagging campaign). Reindex those sites.

1. **Publish verticals & bookmarks tenant-wide.** Move `Search Vertical – Policies – Prod` and

`Bookmark – VPN Access – Prod` from Test to **Published**; add the remaining bookmarks from the query inventory.

1. **Re-validate at tenant scope** with `Test-Findability.ps1` against a broader query set, and monitor with

the health helper:

```
.\Get-SearchHealthScope.ps1 -AdminUrl https://<tenant>-admin.sharepoint.com `
    -OutputPath .\Search-Health-Tenant.json
```

1. **Operationalize.** Schedule the health helper monthly; keep the refinable register current; require new

content types to carry the standard site columns so new content is findable by default.

Gate 6: tenant-wide validation passes and the health report shows the mappings, sources, verticals, and bookmarks present and consistent.

Rollback (documented, every tenant-level change)

- **Schema:** re-import the baseline — `Set-PnPSearchConfiguration -Scope Subscription -Path .\SearchConfig-Baseline.xml`

— or, in the Search admin center, remove the added mapping/alias from the managed property. Reindex.

- **Result source / vertical:** delete `Result Source – Policies – Prod` and `Search Vertical – Policies – Prod`

in the admin centers (or set the vertical to disabled).

- **Bookmark / promoted result:** set `Bookmark – VPN Access – Prod` to Draft/Unpublished (or delete the

query rule promoted result on the site).

- **Content types:** unpublish the content type; existing tagged content and its crawled properties remain but

stop propagating. Reindex latency applies to every rollback — allow propagation time before declaring the revert complete.

5. Testing & Validation — SharePoint Online Search & Information Architecture

Run these after Stage 3–4 on the pilot (Gate 5) and again after Stage 6 tenant-wide (Gate 6). Each test has an explicit **pass/fail**. Automate the query tests with `Test-Findability.ps1`; do the experience checks (verticals, bookmarks) in the UI. Remember **reindex latency** — if a test fails right after a schema change, confirm the reindex/crawl has completed before treating it as a real failure.

Test 1 — Metadata & content types are applied and populated

- **How:** Run `Get-SearchIAAssessment.ps1`; open a pilot library and confirm items carry Department/DocType/Owner.
- **Pass:** `Content Type – Policy Document – Prod` is present and published; pilot documents have

Department and DocType values populated.

- **Fail:** content type missing/unpublished, or columns empty — return to Stage 2/3.1.

Test 2 — Crawled properties exist

• **How:** Search admin center → Manage Search Schema → Crawled Properties (search `Department`), or read the

assessment output.

• **Pass:** crawled properties for the Department and DocType columns (`ows_*`) exist.

• **Fail:** absent → the column was not populated or the reindex has not completed (Stage 3.2–3.3).

Test 3 — Managed property mappings & flags

• **How:** `Test-Findability.ps1` reads the schema; or open `RefinableString00/01` in the Search admin center.

• **Pass:** `Department` (alias on `RefinableString00`) and `DocType` (on `RefinableString01`) are mapped to

the correct crawled properties with **Queryable + Retrievable + Refinable** set.

• **Fail:** wrong crawled property, missing alias, or missing flag → re-run `Set-ManagedPropertyMapping.ps1`.

Test 4 — Queryable: property-targeted query works

• **How:** In pilot search (or `Submit-PnPSearchQuery -Query 'DocType:Policy'`), query `Department:Finance`.

• **Pass:** results are restricted to items whose Department term is Finance.

• **Fail:** query ignored / returns everything → managed property not Queryable, or alias wrong (Test 3).

Test 5 — Retrievable: value returned in results

• **How:** `Test-Findability.ps1` requests `-SelectProperties "Title,Department,DocType,Path"`; or inspect a

result card configured to show DocType.

• **Pass:** the `Department`/`DocType` values are returned for matching items.

• **Fail:** blank values → managed property not Retrievable (Test 3).

Test 6 — Refinable: refiners populate

• **How:** On the pilot results page, add a **Department**/**DocType** refiner (or check the refiner data in

the query result groups).

• **Pass:** the refiner lists the distinct term values and filtering by one narrows results.

• **Fail:** no refiner values → property not Refinable, or content not yet reindexed after mapping (Stage 3.5).

Test 7 — Known-item queries return the expected top result

• **How:** `Test-Findability.ps1 -QueryCsv .\test-queries.csv` runs each priority query and compares the top

result to the expected URL.

• **Pass:** every priority known-item query returns its expected page/document on the first page.

• **Fail:** expected result missing/buried → check content-type scoping, add a bookmark/promoted result, or

fix metadata on the target.

Test 8 — Result source scopes correctly

• **How:** Query through `Result Source – Policies – Prod` (admin center test, or the vertical).

• **Pass:** only policy content is returned; non-policy content is excluded.

• **Fail:** unrelated content appears → the query transform (content-type ID / path) is too broad (Stage 4.1).

Test 9 — Vertical appears and works

• **How:** Perform a search in the Microsoft Search box; select the **Policies** tab

(`Search Vertical – Policies – Prod`).

• **Pass:** the Policies tab appears and lists only policy content.

• **Fail:** tab missing/empty → vertical disabled, wrong content source, or query scope wrong (Stage 4.2).

Test 10 — Bookmark / promoted result fires

• **How:** Search the bookmark keyword (e.g. `vpn`).

• **Pass:** `Bookmark – VPN Access – Prod` appears above organic results for its keywords.

• **Fail:** absent → bookmark in Draft/Test, keyword mismatch, or not yet propagated (allow a few minutes).

Test 11 — No regression in general search

• **How:** Run a set of common general queries before and after the change on the pilot.

• **Pass:** general relevance is unchanged or improved; nothing previously found is now missing.

• **Fail:** general results degrade → a tenant mapping/alias collision (check the refinable register), roll back.

Test 12 — Rollback verified (pre-expansion)

• **How:** In a non-production or pilot check, confirm the baseline re-import restores prior schema

(`Set-PnPSearchConfiguration -Scope Subscription -Path .\SearchConfig-Baseline.xml`) and vertical/bookmark removal works.

- **Pass:** rollback restores the pre-change state after reindex.
- **Fail:** baseline import errors → re-capture a clean baseline before proceeding to tenant scope.

Overall pass criteria (Gate 5 / Gate 6): Tests 1–11 pass on the pilot before expansion; Tests 3–11 pass tenant-wide after expansion; the validation report `Findability Validation – Pilot – Validation` is saved with the change record.

6. Troubleshooting — SharePoint Online Search & Information Architecture

Common failure modes and resolutions. The single most frequent root cause is **reindex latency** — schema and metadata changes do not appear until the affected content is re-crawled. Confirm the crawl has completed before treating any of the below as a real failure.

A refiner shows no values (empty refiner)

- **Symptom:** a Department/DocType refiner appears but lists nothing, or the refiner never appears.
- **Likely causes / fixes:**

1. **Property not Refinable** — open the managed property (`RefinableString00`) in the Search admin

center and confirm **Refinable = Yes**; re-run `Set-ManagedPropertyMapping.ps1` if not.

1. **Content not reindexed** — the mapping was set but the library/site has not been re-crawled. Force

Reindex document library / Reindex site and wait for the crawl.

1. **Metadata empty** — the documents have no Department/DocType value, so there is nothing to refine on.

Populate the columns (Test 1) then reindex.

A property-targeted query is ignored (`Department:Finance` returns everything)

- **Symptom:** scoping a query by a managed property has no effect.
- **Fix:** the property is **not Queryable** or the **alias** doesn't match. Confirm the alias

(`Department`) is set on the mapped managed property and Queryable is checked; then query by the alias.

A managed property returns blank values

- **Symptom:** results show no value for DocType even though the item is tagged.
- **Fix:** the property is **not Retrievable**. Set Retrievable on the managed property and reindex; verify

with `Test-Findability.ps1 -SelectProperties`.

The crawled property never appears in the schema

- **Symptom:** after mapping time, the `ows_Department` crawled property isn't listed.
- **Likely causes / fixes:**

1. **Column not populated on any crawled item** — a crawled property is created only once content with

that column populated is crawled. Tag at least one item and reindex.

1. **Reindex not completed** — allow the crawl to finish; re-run `Get-SearchIAAssessment.ps1`.
2. **Wrong column type** — a Managed Metadata column surfaces differently from a text column; confirm the

exact crawled-property name in the admin center before mapping (don't guess the `ows_*` name).

Mapping change "isn't taking effect"

- **Symptom:** you set a mapping but nothing changed.
- **Fix:** mappings apply only to content indexed **after** the change. **Reindex** the affected libraries/

sites. At tenant scope this propagates over hours-to-days — communicate the window; don't re-apply the mapping repeatedly (that won't speed it up).

Refiners/results regressed tenant-wide after a change

- **Symptom:** general search got worse, or an unrelated team's refiner broke.
- **Likely cause:** an **alias/refinable collision** — two different crawled properties mapped to the same

refinable slot/alias. Check the **refinable-allocation register**, remove the conflicting mapping, and **roll back** with the baseline (`Set-PnPSearchConfiguration -Scope Subscription -Path SearchConfig-Baseline.xml`). Reindex.

The Policies vertical (tab) doesn't appear

- **Symptom:** users don't see `Search Vertical – Policies – Prod`.
- **Likely causes / fixes:**

1. **Vertical disabled / still in draft** — enable/publish it in the Search & Intelligence admin center.
2. **Wrong surface** — Microsoft Search verticals appear on the Microsoft Search results page (app bar /

Office.com / SharePoint home), **not** a classic site search box. Confirm where users start.

1. **Query scope returns nothing** — the vertical's query (e.g. `ContentType:"Policy Document"`) matches no

indexed content yet; confirm content is tagged and indexed (Tests 1–2).

The result source returns unrelated content

- **Symptom:** `Result Source – Policies – Prod` includes non-policy items.

• **Fix:** the **query transform** is too broad. Tighten it with the exact content-type ID

(`ContentTypeId:0x0101...*`) and/or the policies hub **path**; test in the admin center's result-source test pane.

A bookmark / promoted result doesn't fire

• **Symptom:** searching the keyword doesn't surface `Bookmark – VPN Access – Prod`.

• **Likely causes / fixes:**

1. **State is Draft/Test** — set the bookmark to **Published** (or ensure your account is in the Test

audience).

1. **Keyword mismatch** — add the exact terms users type (`vpn`, `remote access`); keywords are matched,

not fuzzy-guessed.

1. **Propagation delay** — allow a few minutes after publishing.

2. **Classic vs Microsoft Search** — a site **query-rule promoted result** only fires on that site's

classic search, not tenant Microsoft Search; use a Bookmark for the tenant experience.

`Connect-PnPOnline` / `Connect-SPOService` fails or the schema export errors

• **Symptom:** connection or `Get-PnPSearchConfiguration` denied.

• **Likely causes / fixes:**

1. **Insufficient role** — you need Search Administrator / SharePoint Administrator; confirm the role.

2. **PnP app not registered** — some tenants require a registered PnP Entra app for interactive auth;

register it (`Register-PnPManagementShellAccess`) or use the app registration your org mandates.

1. **Wrong cloud URL** — Government clouds use `*-admin.sharepoint.us`; add `-AzureEnvironment USGovernment`

where required. **Verify in your tenant.**

Content types stopped propagating to sites

• **Symptom:** new sites don't get the standard content types.

• **Fix:** confirm the content type is **published** (content type gallery: Publish; Content Type Hub:

Manage publishing → Publish) and, for the hub, that the content-type publishing timer job has run. Run `Get-SearchHealthScope.ps1` to list sites missing the content type.

When to escalate to Microsoft

If crawled properties still don't appear or refiners stay empty long after confirmed reindex (days) with populated metadata and correct mappings, open a Microsoft support case with the site URL, the

managed/crawled property names, and the timeline — this can indicate a service-side index issue rather than a configuration error.

7. Best Practices — SharePoint Online Search & Information Architecture

Durable practices for sustained findability. These outlive the initial project; treat them as the operating standard.

Information architecture & metadata

- **Metadata before schema before experience.** Never tune the search schema for content that has no consistent metadata — you will refine on nothing. Fix the IA first.
- **Keep the controlled vocabulary small and governed.** A handful of high-value term sets (`Term Set – Department – Prod`, `Term Set – DocType – Prod`) used everywhere beats dozens of local, free-text columns. Assign a term-store owner and a change process.
- **Publish enterprise content types centrally.** Use the content type gallery / hub so every library produces the same crawled properties. Require new site templates to inherit the standard content types so new content is findable by default.
- **Make key metadata required (or defaulted)** on high-value libraries so documents can't be saved untagged. Untagged content is the number-one cause of empty refiners.

Search schema discipline

- **Reuse the built-in refinable properties** (`RefinableString/Date/Int/Decimal`) with **aliases** — don't proliferate custom managed properties. Maintain the **refinable-allocation register** so two teams never map different metadata to the same alias (a silent, tenant-wide corruption).
- **Set only the flags you need.** Refinable/Sortable have index cost; mark a property Refinable only if a refiner/sort actually uses it. Retrievable + Queryable is enough for many properties.
- **Prefer site-collection scope for narrow needs; reserve tenant scope for genuinely global metadata.** Tenant-level changes reindex everything and affect every site.
- **Always keep a current schema baseline export** (`Get-PnPSearchConfiguration -Scope Subscription`) so any change is reversible.

Change management & reindex

- **Pilot-first, always.** Prove mappings, sources, verticals, and bookmarks on `Pilot Site – Findability – Pilot` and pass the gates before tenant-wide schema changes.

- **Respect reindex latency.** After a mapping change, refiners and returned values populate only as content

is re-crawled — hours to days at tenant scale. Communicate the propagation window; don't declare success (or failure) before the crawl completes.

- **Make tenant-level schema changes in a change window** with a captured baseline and a communicated

rollback.

Search experience

- **Scope result sources tightly** with content-type IDs and/or hub paths so a vertical returns only what it

should. A "Policies" tab that also shows random documents erodes trust fast.

- **Bookmarks for known-item queries, verticals for exploration.** Pin authoritative answers

(`Bookmark – VPN Access – Prod`) for high-volume, single-answer searches; use verticals for "browse a category" needs.

- **Curate bookmarks from real query data.** Seed from the service desk's top "can't find X" tickets and

from Microsoft Search usage insights; retire bookmarks whose targets move or die.

- **Confirm where users actually start** (Microsoft Search box in the app bar / Office.com / SharePoint home

vs a hub's own search) and configure that surface — the vertical/bookmark model differs between Microsoft Search and classic site search.

Governance & sustainment

- **Run the health helper on a schedule** (`Get-SearchHealthScope.ps1`, monthly) to catch mapping drift,

broken bookmark targets, and content types that stopped publishing.

- **Review the query inventory quarterly.** Findability needs change as the org and content change; refresh

test queries and re-validate.

- **Keep documentation and the refinable register versioned** with the schema baseline, so the next admin

inherits the map, not a mystery.

Safety, boundaries & compliance

- **Automation is safe-by-default.** Scripts are read-only or `-WhatIf` unless an explicit `-Commit`/`-Apply`

switch is passed; use modern-auth `Connect-*`; never store credentials or secrets in code.

• **Search reflects permissions — it does not grant them.** Making content findable does not change who can

open it; security trimming still applies. Do not "fix findability" by loosening permissions.

• **Defer compliance/records determinations.** Which content is a record, its retention label, and any

information-barrier restriction are decisions for the compliance/records/legal functions; this solution makes content findable, it does not decide what must be retained or restricted.

• **Verify plan-/cloud-dependent behavior in your tenant.** Content-type gallery vs Content Type Hub,

Microsoft Search feature availability, and Government-cloud URLs (`*.sharepoint.us`, `*.office365.us`) vary — confirm, don't assume.

8. Deliverables Summary — SharePoint Online Search & Information Architecture

Documentation

• **01 — Solution Overview:** how SPO/Microsoft Search works (crawling, the search schema, result

sources/verticals, promoted results) and the IA that feeds it; the reference design, limitations, naming convention, and deployment model.

• **02 — Requirements:** solution-specific prerequisites (Search Administrator access, term-store rights,

content-type publishing surface, findability intent + query inventory, refinable-property register, pilot scope, reindex/change window) building on the shared SharePoint global requirements.

• **03 — Implementation:** the ordered runbook — Assess → Design → Configure metadata/content types →

Configure search schema → Configure result sources/verticals/bookmarks → Pilot → Validate → Expand — with every artifact named and explicit pass/fail gates plus a documented rollback.

• **05 — Testing & Validation:** twelve tests (metadata, crawled/managed properties, queryable/retrievable/

refinable, known-item queries, result source, vertical, bookmark, no-regression, rollback) each with pass/fail.

• **06 — Troubleshooting:** common failure modes (empty refiners, mapping not taking effect, crawled property

missing, vertical/bookmark not showing, alias collisions) and resolutions.

• **07 — Best Practices:** durable practices for IA, schema discipline, change/reindex management, the search

experience, sustainment, and safety/compliance boundaries.

• **Executive Summary:** outcome, approach, and why it is low-risk.

PowerShell scripts (5) — SharePoint Online Management Shell / PnP / Graph, safe-by-default

1. `Get-SearchIAAssessment.ps1` — *Discovery (read-only).* Enumerates term sets, published content types,

site columns, and the current managed-property → crawled-property mappings with their flags; captures the tenant search-config **rollback baseline**. Produces `Findability Assessment – Tenant – Discovery`.

1. `Set-ManagedPropertyMapping.ps1` — *Configuration helper (`-WhatIf` by default).* Maps crawled →

refinable managed properties with alias and searchable/queryable/retrievable/refinable flags from a CSV; dry-runs unless `-Commit` is passed; always exports a baseline first and re-imports the edited schema at the chosen scope.

1. `Set-SearchSourceVertical.ps1` — *Experience config helper (`-WhatIf` by default).* Creates scoped

result sources and enumerates the **verticals** and **promoted results / bookmarks** to configure in the Microsoft Search & Intelligence admin center; dry-runs unless `-Commit` is passed.

1. `Test-Findability.ps1` — *Validation (read-only).* Runs the test-query inventory with

`Submit-PnPSearchQuery`, checks each returns its expected top result and required managed properties, and writes `Findability Validation – Pilot – Validation` with a pass count (Gate 5 / Gate 6).

1. `Get-SearchHealthScope.ps1` — *Scope/health helper (read-only).* Reports content-type/metadata

coverage across sites, confirms expected managed-property aliases exist in the tenant schema, and checks bookmark-target reachability — for scheduled monitoring and drift detection.

All scripts use modern-auth `Connect-*` (no stored credentials or secrets), default to read-only or `-WhatIf`, and require an explicit `-Commit`/`-Apply` switch before any change.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

Get-SearchHealthScope.ps1

Get-SearchIAAssessment.ps1

Set-ManagedPropertyMapping.ps1

Set-SearchSourceVertical.ps1

Test-Findability.ps1